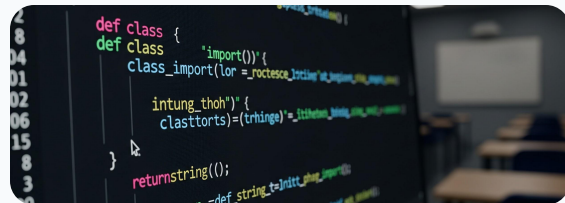


Project Overview: Student Performance Analyzer

Your Mission

Act as a **data analyst for a school**. Analyze student performance data using Python to drive meaningful insights.



Skills to Apply

- Variables
- Lists & Dictionaries
- Loops
- Conditionals
- Functions

Key Outcomes



Use Python to analyze data and make meaningful conclusions.

The Dataset

Defining the Data Structure

We will start with a small dataset of students represented as a **list of dictionaries**.

Key Concepts:

- **List:** An ordered collection of items.
- **Dictionary:** Key-value pairs for structured attributes.
- **Data Fields:** Each student has a name, grade, and attendance record.

students.py

```
students = [  
    {"name": "Alice", "grade": 85, "attendance": 90},  
    {"name": "Bob", "grade": 65, "attendance": 75},  
    {"name": "Charlie", "grade": 92, "attendance": 88},  
    {"name": "Diana", "grade": 70, "attendance": 95},  
    {"name": "Ethan", "grade": 78, "attendance": 60},  
    {"name": "Fatima", "grade": 88, "attendance": 82},  
    {"name": "Gabriel", "grade": 58, "attendance": 91},  
    {"name": "Hannah", "grade": 73, "attendance": 79}  
]
```



Python Logic: Exploration & Conditionals

Part 1: Exploring the Data

Loop through the dataset and print each student's name and grade.

Your Task:

- Use a `for` loop
- Access values using dictionary keys

```
for student in students:  
    print(student["name"], student["grade"])
```

Part 2: Pass or Fail

Determine if each student is passing or failing based on their grade.

Rule:

- Passing = grade $\geq$ 70
- Failing = grade $<$ 70

```
for student in students:  
    if student["grade"] >= 70:  
        print(student["name"], "Pass")  
    else:  
        print(student["name"], "Fail")
```

Python Logic: Deeper Analysis & Functions

Part 3: Deeper Analysis

Include attendance in your evaluation logic.

Rule:

- Passing & Engaged = grade $\geq$ 70 AND attendance $\geq$ 80

Your Task:

- Use logical operators (**and**)

```
for student in students:
    if student["grade"] >= 70 and student["attendance"] >= 80:
        print(student["name"], "is passing...")
```

Part 4: Creating a Function

Organize the evaluation logic into a reusable function.

Your Task:

- Write **evaluate_student**
- Return "On Track" or "At Risk"

```
def evaluate_student(student):
    if student["grade"] >= 70 and ...:
        return "On Track"
    else:
        return "At Risk"
```

Python Logic: Implementation & Insights

Part 5: Apply the Function

Use your function on all students.

Your Task:

- Loop through the dataset
- Call your function

```
for student in students:  
    result = evaluate_student(student)  
    print(student["name"], "-", result)
```

Part 6: Data Insight

Now we move into real data analysis.

Your Task:

- Count how many students are "At Risk"

This allows us to identify trends and prioritize support for students who may need additional help.

```
at_risk = 0  
for student in students:  
    if evaluate_student(student) == "At Risk":  
        at_risk += 1  
print("Number of at-risk students:", at_risk)
```

Conclusion & Extensions

Reflection Questions

Answer the following:

- What patterns did you notice in the student data?
- Why might attendance be important in evaluating students?
- How could this program be improved in a real-world setting?

Extension Challenges (Optional)

Try one or more:

- Find the student with the highest grade
- Calculate the average grade
- Add a new student to the dataset
- Ask the user to input student data
- Sort students by grade

Submission Instructions

Final Steps

- Make sure your notebook runs from top to bottom
- Use clear variable names
- Include comments explaining your code
- Ensure your notebook runs from top to bottom

Your Data Journey

This is your **first real data project**.

You are no longer just writing code—you are:

analyzing information and making decisions

That is the foundation of **data science**.